



OPEN Effect of enzyme preparation and extrusion puffing treatment on sorghum straw silage fermentation

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In this study, the effects on silage performance and microbial community of sorghum straw treated with the addition of enzymes (cellulase (CE), xylanase (XE)) and extrusion puffing technology, combined with SEM, XRD, and FTIR techniques, were thoroughly investigated. The results showed that the enzyme preparations, especially xylanase, significantly improved the nutritional value and fermentation efficiency of straw and enhanced the silage effect. Extruding significantly changes the surface structure of the straw, increasing the surface area and porosity, and promoting the attachment of microorganisms. This study not only optimized the sorghum straw silage performance but also provided technical support for the efficient use of straw resources, which is of great significance for the sustainable development of animal husbandry and the resource utilization of agricultural waste.

Keywords Enzyme preparation, Extrusion puffing treatment, Sorghum straw, Silage performance, Microbial community

Abbreviations

AA	Acetic acid
ADF	Acid detergent fiber
CE	Cellulase
CP	Crude protein
DM	Dry matter
EPFM	Extrusion puffing matter
FM	Fresh matter
FTIR	Fourier Transform Infrared Spectrometer
KEGG	Kyoto Encyclopedia of Genes and Genomes
LAB	Lactic acid bacteria
LA	Lactic acid
NDF	Neutral detergent fiber
NH ₃	N-Ammonia nitrogen
PA	Propionic acid
SEM	Scanning electron microscope
WSC	Water-soluble carbohydrate
X	Ray diffraction-XRD
XE	Xylanase

Sorghum (*Sorghum bicolor* (L.) Moench) is a versatile graminaceous annual herbaceous plant that is resistant and nutritious¹. Sorghum is the fifth-largest cereal crop globally and has a wide cultivation area and geographical distribution. Sorghum straw is rich in nutrients such as fiber, proteins, and sugars, which have high feeding value. Still, due to its hard texture, it is rich in lignin, cellulose, and hemicellulose², which makes sorghum straw highly structurally stable and resistant to biodegradation. To enhance the utilization rate of sorghum straw, it is imperative to implement efficacious pretreatment methods to substantially reduce the lignocellulosic content of the straw, thereby diminishing its hardness and stability.

Cellulases and xylanases have significant effects on glucose or xylan released from lignocellulosic biomass³. Cellulase has the function of breaking down cellulose in plant cell walls and converting it into more carbohydrates, thus providing more energy to lactic acid bacteria for fermentation⁴. Xylanase can break down xylan into smaller

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sugar fractions and break down hemicellulose to further release bound nutrients⁵. In several studies, it has been found that enzyme additives have demonstrated significant results in the treatment and utilization of biomass and feedstuffs. Alokika et al.⁶ found that the addition of xylanase to bagasse resulted in a significant increase in glucose and xylose content. Li et al.⁷ found that the addition of cellulase can enhance the fermentation quality of silage, and also inhibit protein degradation and accelerate the rate of fiber decomposition. T.A. Del Valle et al.⁸ found that the addition of xylanase to silage significantly promotes plant cell wall decomposition, reduces NDF content, and improves the digestibility and nutritional value of the feed.

Straw extrusion puffing feed processing technology (straw extrusion puffing for short) is an emerging feed processing technology that utilizes thermal and mechanical effects to treat straw at high temperatures and pressure⁹. After the extrusion puffing treatment of sorghum straw, the fibers were separated into filaments and the surface was rough, which greatly facilitated the attachment and propagation of microorganisms, thus enhancing their degradation capacity and enabling increased feed intake and utilization¹⁰. The treated straw is soft and easy to cut and compact, reduces air retention, improves silage density and capacity, reduces loss and waste, and enhances enzymatic digestion. In addition, extrusion puffing destroys the particle and crystal structure of sorghum straw and also destroys the waxy membrane on the surface of the straw, separates cellulose, hemicellulose, and lignin, promotes the contact of beneficial bacteria with the straw, improves the fermentation efficiency, and ultimately enhances the quality of silage¹¹.

To date, the effects of adding enzymes to silage have been studied more extensively¹². However, the mechanism of the synergistic effect of enzyme preparation and extrusion puffing treatment on improving the silage quality of sorghum straw is not clear. Therefore, this study aimed to evaluate the effect of the addition of cellulase, xylanase, and after extrusion puffing of enzyme preparation on sorghum straw silage by analyzing the silage performance and bacterial community structure and observing the changes in the surface morphology and structure of sorghum straw with the help of SEM, XRD, and FTIR.

Materials and methods

Extrusion pretreatment

The pre-treated sorghum straw passes through the screw extrusion system of the Model 9P-150 C Straw Extruding and Puffing Machine (Liaoning Xianghe Agricultural and Animal Husbandry Industry Co., Ltd.). The extrusion puffing sorghum straw is discharged through the outlet and collected for cooling. The equipment has twin-screw extrusion, heating, and control systems. Through the rotation and propulsion of the screw, the straw is strongly extruded, stirred, and sheared in the barrel. The heating system ensures that the straw is treated at an optimal temperature range of 130–160 °C, facilitating the partial degradation or reconfiguration of components such as cellulose, hemicellulose, and lignin. This process loosens the structure, making it more amenable to microbial decomposition and utilization. Within this temperature spectrum, lignin undergoes melting and thermohydrolysis, with partial disruption of hydrogen bonds and a reduction in crystallinity. Such alterations in lignin help dismantle the compact architecture of the straw, rendering it more porous, thereby enhancing its digestibility and utility while ensuring the safety and stability of the feed.

Ensiling preparation

Sorghum straw at maturity was harvested in the area of Balinzuo Banner, Chifeng City, Inner Mongolia, and the test material was harvested in September–October 2022. Afterwards, the harvested straw was pulverized to 3–5 cm. Some of the sorghum straws were added with enzymes and some of the sorghum straws were extruded and puffed to add enzymes. The experiment was divided into six treatments: (1) no additives added equal amount of distilled water (CK); (2) cellulase (CE); (3) xylanase (XE); (4) extrusion puffing with an equal amount of distilled water (EPCK); and (5) Addition of cellulase (EPCE) after extrusion puffing. (6) Addition of xylanase (EPXE) after extrusion puffing. Both enzyme preparations (powders) were obtained from Inner Mongolia Hohhot Yidu Biotechnology Co. The activity was 5×10^5 U/g DM. In the experiment, the addition of 0.2 g¹³ of dry matter enzyme preparation ensures an enzymatic activity level of 1×10^5 U/kg DM. At this level of activity, the enzymes can efficiently act on the substrate, breaking it down into smaller molecules that are more readily utilized by microorganisms, thereby enhancing the digestibility and utilization rate of the straw. Dissolve the additives in water and evenly spray in the raw materials, 500 g per bag into a polyethylene vacuum bag sealed. Three replicates per treatment. Nutrient composition, fermentation quality, and microbial community of the silage were analyzed after 30 d and 60 d of silage at room temperature.

Determination of chemical composition

The material was placed in an oven at 105 °C for 3 h of killing treatment and then dried in an oven at 65 °C until a constant weight was reached and the dry matter (DM) content was determined. Kjeldahl method¹⁴ was used to determine the crude protein (CP) content of the material. Neutral detergent fiber (NDF) and acid detergent fiber (ADF) contents were determined using an ANKOM fiber analyzer (model: A2000i; manufactured by Anke Borealis Technology Co., Ltd. in Beijing, China) according to Van Soest¹⁵ et al. Anthrone colorimetry was used to assess the content of water-soluble carbohydrates (WSC) in the material¹⁶.

Fermentation quality measurement

10 g of silage sample was mixed with 90 ml of distilled water and the mixture was stirred for 2 min using a tapper (JX-05) to obtain an organic acid liquid. The pH of the liquid was measured using a pH meter (PHS-3 C, Shanghai Precision Scientific Instruments Co., Ltd., China). Concentrations of lactic acid (LA), acetic acid (AA), propionic acid (PA), and butyric acid (BA) were determined using a high-performance liquid chromatography (HPLC) system (model: WatersE2695, Milford, USA). For the separation chromatographic column, the KC-811 column of the RSpak series was used, with a size of 8 mm×300 mm. The mobile phase was a 3 mmol solution

of good quality perchloric acid. During the chromatographic analysis, the flow rate was set at 1 mL/min, the temperature of the column was steadily maintained at 40 °C, and the detection wavelength was set at 210 nm. The sample volume for each injection was 5 µL. Ammonia nitrogen (NH₃-N) was measured using phenol-sodium hypochlorite colorimetric method¹⁷.

Bacterial community sequencing analysis

DNA extraction kit (FastDNATMSpinKitforSoil, MPBiomedicals, USA) was used to extract total DNA from the samples. The quality and concentration of extracted DNA were detected by 1% agarose gel electrophoresis (voltage 5 V/cm for 20 min) and spectrophotometric method (optical density 260/280 nm). The V3-V4 variable region of the bacterial 16S rRNA gene was selected and the extracted DNA template was amplified by PCR using primers 338 F (ACTCCTACGGGGAGGCAGCAG) and 806R (GGACTACHVGGGTWTCTAAT)¹⁸. Its products were purified using a DNA gel recovery purification kit (PCRClean-UpKit, Overwatch, China), and Qubit 4.0 (ThermoFisherScientific, USA) was used for detection and quantification¹⁹, and mixed according to the sequencing requirements of each sample in the appropriate proportions. Initial detection was performed using a 2% agarose gel and sequencing was performed using Illumina's MiseqPE300/NovaSeqPE250 platform.

Bioinformatics analysis

Alpha diversity, Chao1, and Shannon's index were calculated using Mothur²⁰ software (<http://www.mothur.org/wiki/Calculators>), and the Wilcoxon rank-sum test was used for the analysis of between-group differences in alpha diversity; QIIME2 random normalization was used, to calculate beta diversity. The relative proportions of bacterial communities indicate the distribution and abundance of different bacterial taxa in a given environment. Differences in bacterial groups under different enzyme treatments and fermentation times were calculated by weighted UniFrac distances, and similarities in microbial community structure among samples were examined using PCoA analysis (Principal Coordinate Analysis) based on the Bray-Curtis distance algorithm; LEfSe analysis (Linear discriminant analysis effect size) was used to²¹ (<http://huttenhower.sph.harvard.edu/LEfSe>) (LDA > 2, $P < 0.05$) to identify bacterial taxa with significant differences in abundance from phylum to genus level among different groups. Metabolic genes were predicted using PICRUSt based on 16S rRNA data²². The Kyoto Encyclopedia of Genes and Genomes (KEGG) was also used to assign genes to metabolic pathways. Predicted functional characteristics of bacterial communities were taken from the Kyoto Encyclopedia of Genes and Genomes (KEGG; <http://www.genome.jp/kegg/>) via PICRUSt2 (<https://github.com/picrust/picrust2>; version 2.2.0), which, based on the proportion of tagged gene sequences in a sample predicts the function of the bacterial community. Pathwaylevel1, Pathwaylevel2, and Pathwaylevel3 correlation analyses were constructed using Origin2021.

Scanning electron microscopy analysis

Scanning electron microscope (CarlZeissAG, Germany) ZEISSSigma500 was used to observe the morphological structure of sorghum straw, sorghum straw after extrusion puffing, and sorghum straw after the addition of enzyme silage. The samples were coated with gold before any SEM images were captured. The coating thickness was 100–150 Å and an E-1010 vacuum sputterer was used in the coating process. SEM images were taken at 1000 and 2000 magnification using an S4800 field emission SEM instrument at an accelerating voltage of 10 kV.

X-ray diffraction (XRD) analysis

In this experiment, the crystallinity of sorghum straw samples was analyzed using an X-ray powder diffractometer (PANalyticalX'pertPRO, Netherlands). The radiation source was Cu-Kα, the scanning angle range was 10°–60° (2θ), the scanning voltage was 30 kV, the scanning current was 30 mA, and the scanning rate was 2°/min with a step size of 0.02°. The crystallinity index (CrI) was calculated according to the Segal method²³.

Fourier transform infrared spectroscopy (FTIR)

In this study, a Fourier infrared spectrometer (BrukerALPHA, Germany) was used to detect changes in sorghum straw samples before and after treatment. The samples were ground into powder and pressed into microspheres at 30 MPa. The scanning wavelength of each sample was 400–4000 cm⁻¹ and all samples were vacuum dried before analysis. Results were analyzed using OMNIC-8.2 software.

Data analysis

The experimental data were organized using Excel 2019 software and a two-way analysis of variance (ANOVA) was performed on the effects of additives and days of silage on silage fermentation parameters, chemical composition, and microbial α-diversity using SAS 9.4 statistical software. Origin 21.0 software was used to graph the data to represent and analyze the correlation between silage indicators, providing a visual understanding of the experimental results. The results obtained from FTIR analysis were analyzed by OMNIC-8.2 software.

Results

Fermentation characteristics of silage from different treatments of sorghum straw and extrusion puffing sorghum straw silage

The effects of additives and extrusion puffing treatments on the fermentation characteristics of sorghum straw silage are shown in Table 1. For both periods of silage, pH was significantly ($p < 0.05$) higher in the CK and EPCK groups than in the other treatment groups, and pH was higher in all treatments at 60d of silage than at 30d of silage. LA content increased with increasing silage time, which was particularly pronounced in the XE group and the treatment groups after extrusion puffing. AA content tended to increase but remained within reasonable limits. The LA and AA contents were higher in the extrusion puffing treatment groups than in the sorghum straw

Items	Ensiling days	Ensiling days						SEM	P-Value		
		CK	CE	XE	EPCK	EPCE	EPXE		T	D	T×D
pH	30	3.60 ^{Ba}	3.52 ^{Bb}	3.55 ^{Bb}	3.60 ^{Ba}	3.38 ^{Bc}	3.55 ^{Bb}	0.012	< 0.0001	< 0.0001	< 0.0001
	60	3.87 ^{Ab}	3.83 ^{Ad}	3.82 ^{Ad}	3.90 ^{Aa}	3.85 ^{Ac}	3.85 ^{Ac}	0.005			
Lactic acid (g/kg DM)	30	15.90 ^{Aa}	8.45 ^{Ab}	11.45 ^{Ab}	8.98 ^{Bb}	16.09 ^{Aa}	28.17 ^{Aa}	14.838	< 0.0001	0.0561	0.0108
	60	6.77 ^{Ad}	5.44 ^{Ad}	15.10 ^{Ac}	26.71 ^{Ab}	17.31 ^{Ac}	44.04 ^{Aa}	19.228			
Acetic acid (g/kg DM)	30	0.84 ^{Bb}	0.52 ^{Cb}	0.74 ^{Bb}	0.41 ^{Cc}	0.67 ^{Bb}	2.09 ^{Aa}	1.448	< 0.0001	0.0002	< 0.0001
	60	0.49 ^{Cc}	0.56 ^{Bb}	1.01 ^{Cb}	0.70 ^{Bb}	0.74 ^{Bb}	2.75 ^{Aa}	1.098			
NH ₃ -N (g/kg DM)	30	0.22 ^{Bd}	0.27 ^{Aa}	0.25 ^{Bb}	0.23 ^{Bc}	0.22 ^{Bd}	0.24 ^{Bb}	0.238	< 0.0001	< 0.0001	0.0146
	60	0.25 ^{Ad}	0.28 ^{Aa}	0.29 ^{Aa}	0.27 ^{Ab}	0.25 ^{Ad}	0.27 ^{Ac}	0.268			

Table 1. Fermentation characteristics of silage from different treatments of sorghum straw and extrusion puffing sorghum straw. DM, dry matter; LA, lactic acid; AA, acetic acid; PA, propionic acid; NH₃-N, ammoniacal nitrogen; CK, control without additives; CE, cellulase; XE, xylanase; EPCK, extrusion puffing control without additives; EPCE, extrusion puffing with added cellulase; EPXE, extrusion puffing with added xylanase; Different lower-case letters denote the same number of days of silage with Different capital letters indicate significant differences between treatments ($p < 0.05$); Different capital letters indicate significant differences between different days of silage for the same treatment ($p < 0.05$); Same letters indicate not significant ($P > 0.05$); ND, not detected; SEM, standard error of the mean; T, treatment; D, days of silage; T×D, interaction between treatment and days of silage.

Items	Ensiling Treatments	
	FM	EPFM
DM	37.12%	40.42%
NDF (% DM)	67.34%	67.79%
ADF (% DM)	47.71%	41.83%
WSC (% DM)	4.81%	6.42%

Table 2. Chemical composition of sorghum straw feedstock and extrusion puffing sorghum straw feedstock. DM, dry matter; NDF, neutral detergent fiber; ADF, acid detergent fiber; WSC, water-soluble carbohydrates; FM, the feedstock of sorghum straw; EPFM, extrusion puffing sorghum straw feedstock.

silage treatment groups. BA was not detected in all treatment groups. NH₃-N content was higher in all treatment groups than in CK control and EPCK control during both silage periods.

Chemical composition of sorghum straw and extruded puffed sorghum straw in different treatments

The chemical composition of sorghum straw feedstock and extrusion puffing sorghum straw feedstock is shown in Table 2. The chemical composition of silage is shown in Table 3. The dry matter content of the extrusion puffing sorghum straw was 40.42%, which was a significant improvement compared to the 37.12% of the raw sorghum straw. Different treatments had significant effects on DM, CP, NDF, ADF, and WSC ($p < 0.05$). The DM content and CP content increased significantly ($p < 0.05$) in all treatment groups with the extension puffing of silage time, with the most significant increase in CP content in the CE and EPCE groups. During the two periods of silage, ADF content was significantly reduced in the CK, CE, and XE treatment groups compared to sorghum straw feedstock, but the ADF content of the extrusion puffing treatment group increased in comparison to the control group, and with the extension of time, the ADF content of the silage treatment group increased, while there were different degrees of decreases in the extrusion puffing treatment group. Similarly, there was a trend of increasing NDF content in all treatment groups compared to the control group. WSC content was significantly ($p < 0.05$) reduced in all treatment groups as the number of days of silage increased.

Bacterial diversity in different treatments of sorghum straw and extrusion puffing sorghum straw silage

The alpha diversity of sorghum straw silage as well as extrusion puffing sorghum straw silage feeds is shown in Table 4, demonstrating the diversity of microbial communities under different treatments. The bacterial coverage of all groups reached 0.99, indicating a high degree of reliability of the bacterial community analysis at this sequencing depth. Silage time significantly ($p < 0.05$) affected Ace, Chao1, Shannon, and Simpson indices. The Ace, Chao1, Shannon, and Simpson values were significantly ($p < 0.05$) lower in the silage after extrusion puffing treatment for 30d and 60d of silage compared to the three treatment groups for sorghum straw silage. In addition, the Ace index and Chao1 index of the cellulase-added treatment group presented lower levels when compared to the CK group without any enzyme addition.

Items	Ensiling days	Ensiling days						SEM	P-Value		
		CK	CE	XE	EPCK	EPCE	EPXE		T	D	T×D
DM (% FM)	30	52.80 ^{Ba}	46.20 ^{Bc}	47.22 ^{Bc}	50.23 ^{Bb}	50.22 ^{Bb}	50.66 ^{Bb}	0.323	<0.0001	<0.0001	<0.0001
	60	64.40 ^{Aa}	52.52 ^{Ae}	53.36 ^{Ad}	59.15 ^{Ab}	54.74 ^{Ad}	57.10 ^{Ac}	0.484			
CP (% DM)	30	13.15 ^{Bb}	13.33 ^{Bb}	13.01 ^{Ab}	12.83 ^{Bb}	15.57 ^{Aa}	12.60 ^{Ac}	0.173	<0.0001	<0.0001	0.0005
	60	14.83 ^{Aa}	15.97 ^{Aa}	13.37 ^{Ac}	14.21 ^{Ab}	15.60 ^{Aa}	12.95 ^{Ad}	0.361			
ADF (% DM)	30	42.82 ^{Bbc}	43.76 ^{Ab}	41.17 ^{Ac}	43.66 ^{Ab}	50.29 ^{Aa}	44.35 ^{Ab}	0.844	0.8792	0.0001	0.0001
	60	46.66 ^{Aa}	44.70 ^{Aa}	42.31 ^{Ab}	46.33 ^{Aa}	44.06 ^{Ba}	42.45 ^{Ab}	0.754			
NDF (% DM)	30	64.47 ^{Ba}	65.51 ^{Aa}	60.06 ^{Ac}	65.29 ^{Ba}	64.60 ^{Aa}	62.23 ^{Ab}	0.372	<0.0001	<0.0001	0.0001
	60	66.96 ^{Ab}	64.07 ^{Bc}	60.53 ^{Ae}	69.70 ^{Aa}	65.56 ^{Ab}	63.53 ^{Ad}	0.383			
WSC (% DM)	30	2.43 ^{Ac}	2.38 ^{Ac}	8.18 ^{Ab}	7.46 ^{Ab}	7.63 ^{Ab}	10.31 ^{Aa}	0.314	<0.0001	<0.0001	<0.0001
	60	0.42 ^{Bd}	0.55 ^{Bd}	1.21 ^{Bc}	1.37 ^{Bc}	1.69 ^{Bb}	4.90 ^{Ba}	0.068			

Table 3. Chemical composition of sorghum straw and extrusion puffing sorghum straw silage in different treatments. FM, fresh material; DM, dry matter; CP, crude protein; NDF, neutral detergent fiber; ADF, acid detergent fiber; WSC, water-soluble carbohydrates; CK, control without additives; CE, cellulase, XE, xylanase; EPCK, control without additives after extrusion puffing; EPCE, cellulase added after extrusion puffing; EPXE, extrusion puffing with the addition of xylanase; different lowercase letters indicate significant differences between treatments for the same number of silage days ($p < 0.05$); different uppercase letters indicate significant differences between the same treatment for different number of silage days ($p < 0.05$); same letters indicate non-significant ($p > 0.05$); ND, not detected; SEM, standard error of the mean; T, treatment; D, number of silage days; T×D, interaction of treatment and silage days interaction.

Items	Ensiling days	Ensiling treatments						SEM	P-Value		
		CK	CE	XE	EPCK	EPCE	EPXE		T	D	T×D
Ace	30	131.37 ^{Aab}	124.97 ^{Bab}	170.83 ^{Aa}	52.60 ^{Bc}	73.23 ^{Abc}	60.80 ^{Bbc}	21.919	<0.0001	0.3396	0.0002
	60	191.78 ^{Abc}	358.54 ^{Aa}	259.64 ^{Ab}	171.27 ^{Abc}	137.08 ^{Ac}	117.4 ^{Ac}				
Chao1	30	130.65 ^{Aab}	124.03 ^{Bab}	169.55 ^{Aa}	51.70 ^{Bc}	73.23 ^{Abc}	60.31 ^{Bbc}	22.075	<0.0001	<0.0001	0.0196
	60	191.24 ^{Abc}	359.82 ^{Aa}	259.42 ^{Ab}	170.58 ^{Abc}	135.99 ^{Ac}	117.18 ^{Ac}				
Shannon	30	1.33 ^{Aa}	1.60 ^{Aa}	1.40 ^{Aa}	0.66 ^{Bb}	0.79 ^{Bb}	0.82 ^{Ab}	0.113	<0.0001	<0.0001	0.2435
	60	1.46 ^{Abc}	2.24 ^{Aa}	1.94 ^{Aab}	1.38 ^{Ac}	1.13 ^{Ac}	1.03 ^{Ac}				
Simpson	30	0.47 ^{Ab}	0.32 ^{Ac}	0.45 ^{Ab}	0.64 ^{Aa}	0.60 ^{Aa}	0.58 ^{Aa}	0.034	<0.0001	0.4647	0.0854
	60	0.46 ^{Ab}	0.29 ^{Ac}	0.37 ^{Abc}	0.51 ^{Bb}	0.65 ^{Aa}	0.68 ^{Aa}				
Coverage	30	99.99 ^{Aa}	99.99 ^{Aa}	99.99 ^{Ab}	99.99 ^{Aa}	99.99 ^{Aa}	99.99 ^{Aa}	0.000	0.0518	0.4403	0.0426
	60	99.99 ^{Aab}	99.99 ^{Aa}	99.99 ^{Aab}	99.99 ^{Aab}	99.99 ^{Aab}	99.99 ^{Aa}				

Table 4. Effect of additives and days of silage on alpha diversity of sorghum straw silage bacteria. CK, control without additives; CE, cellulase, XE, xylanase; EPCK, control without additives after extrusion puffing; EPCE, cellulase added after extrusion puffing; EPXE, xylanase added after extrusion puffing; different lowercase letters indicate significant differences between treatments at the same number of days of silage ($p < 0.05$); different capital letters indicate significant differences between the same treatments at different number of days of silage significant differences among treatments ($p < 0.05$); the same letter indicates non-significant ($p > 0.05$); ND, not detected; SEM, standard error of the mean; T, treatment; D, days of silage; T×D, interaction between treatment and days of silage.

Changes in microbial communities in silage were further verified using beta diversity by principal coordinate analysis (PCoA), as shown in Fig. 1. Figure 1a shows the PCoA plot at 30 days of silage, where the principal coordinate 1 (PCo1) and principal coordinate 2 (PCo2) were 56.86% and 11.25%, respectively. In Fig. 1b presenting the PCoA plot at 60 days of silage, the principal coordinate 1 (PCo1) and principal coordinate 2 (PCo2) were 42.25% and 23.61%, respectively. The bacterial communities of the raw material and control groups were significantly separated from the rest of the treatment groups for both silage periods. At 30 days of silage, the silage samples from different treatments did not show significant differences between them, and the six groups of silage samples showed tighter aggregation in the distribution of bacterial flora. With the prolongation of silage time, the CK, CE, and XE groups showed a clear separation in the bacterial colony structure from the three treatment groups of extrusion puffing.

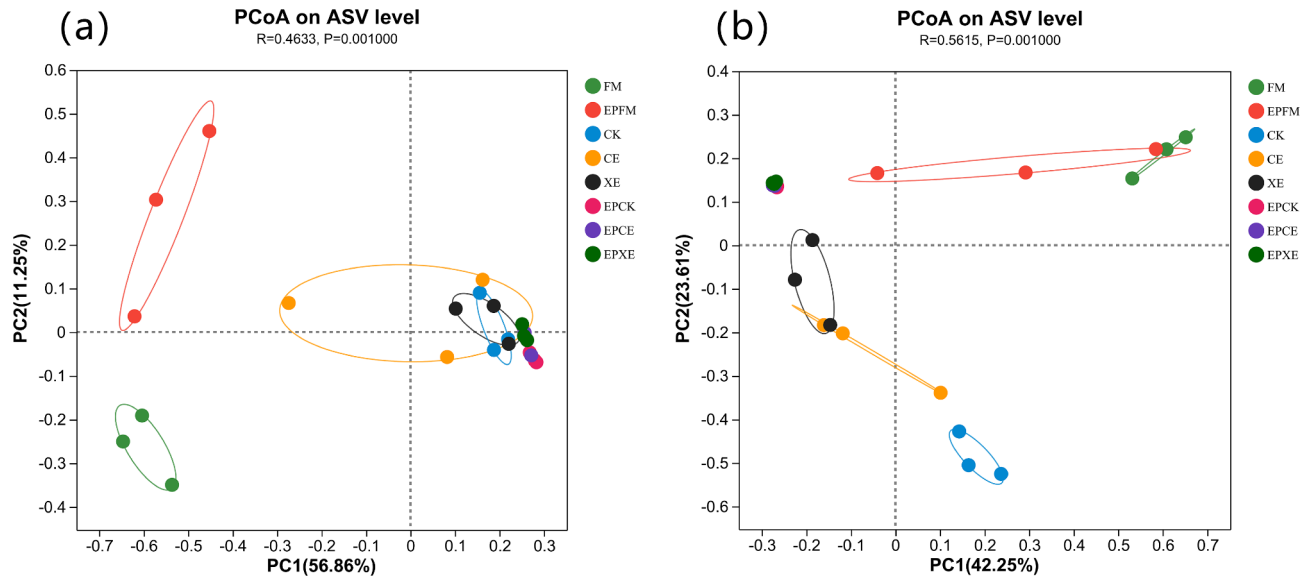


Fig. 1. **a** Representation of the PCoA plot at 30 days of silage. **b** Representation of PCoA plot for 60 days of silage. Differences in bacterial populations under different enzyme treatments and fermentation times were calculated by weighted UniFrac distances, and coordinates were calculated using principal coordinate analysis (PCoA). FM, fresh material; EPFM, extrusion puffing material; CK, additive-free control; CE, cellulase; XE, xylanase; EPCK, extrusion puffing additive-free control; EPCE, extrusion puffing with cellulase; EPXE, extrusion puffing with xylanase.

Community composition of sorghum straw and extrusion puffing straw silage with different additives

Figure 2 illustrates the changes in bacterial flora at the bacterial phylum and genus level for each treatment group. In FM, Proteobacteria (91.22%) and Bacteroidota (6.67%) were the main dominant phyla. Proteobacteria (81.05%) and Firmicutes (12.43%) dominated the EPFM. Figure 2a shows the bacterial gate levels of sorghum straw and extrusion puffing sorghum straw silage at 30 days of silage with different treatments. The dominant phyla in all treatment groups changed significantly after silage, with the originally dominant phyla Proteobacteria and Bacteroidota being replaced by Firmicutes. Figure 2c shows the level of bacterial gates at 60 days of silage. Firmicutes abundance decreased and Proteobacteria abundance increased at 60 days of silage. Firmicutes in the XE group decreased from 83.8–37.66%. Proteobacteria abundance was 72.72%, 85.45%, and 85.64% in EPCK, EPCE and EPXE groups, respectively. As the fermentation process continued, silage diversity gradually decreased at the microbial genus level. At 30 d of silage (Fig. 2b), the CK and XE groups were dominated by *Lactiplantibacillus*, an OTU from an unclassified genus belonging to the family *Alcaligenaceae*, and an OTU from an unclassified genus belonging to the family *Erwiniaceae* predominated. The CE group is dominated by *Lactiplantibacillus*, an OTU from an unclassified genus belonging to the family *Enterobacteriaceae* and *Lentilactobacillus*. The three treatment groups after extrusion puffing were dominated by *Lactiplantibacillus* and *Alcaligenaceae*. At 60 d of silage (Fig. 2d), *Alcaligenaceae* dominated the treatment groups. The six treatment groups were dominated by *Alcaligenaceae*, *Lactobacillus*, and *Lactiplantibacillus*.

Figure 3a shows the one-way ANOVA at the gate level of sorghum straw feedstock with each silage treatment group for 30 days of silage, where FM and EPFM differed significantly ($p < 0.05$) from each treatment group in the Firmicutes and Proteobacteria. Figure 3b shows a one-way ANOVA at the genus level for 30 days of silage for the genera *Lactiplantibacillus*, an OTU from an unclassified genus belonging to the family *Eewiniaceae*, and *Pantoea*, an OTU from an unclassified genus belonging to the family *Enterobacteriaceae*, *Lentilactobacillus*, and an OTU from an unclassified genus belonging to the family *Yersiniaceae* differed significantly ($p < 0.05$) among the FM, EPFM and silage groups. All three treatment groups after extrusion puffing decreased the abundance of most genera and increased the abundance of lactic acid bacteria. The one-way ANOVA at the 60-day silage gate level is shown in Fig. 3c, which showed that there were significant differences ($p < 0.05$) in the microbial composition of the Firmicutes and Proteobacteria between both FM and EPFM and silage treatment groups. One-way ANOVA at the genus level after 60 days of silage (Fig. 3d) revealed that each treatment group had a significant difference between the genera *Alcaligenaceae*, *Lactobacillus*, *Enterobacteriaceae*, *Pantoea*, *Lactiplantibacillus*, *Pseudomonas*, and an OTU from an unclassified genus belonging to the family *Bacillaceae* also differed significantly in abundance ($p < 0.05$). Of particular note was the increased abundance of *Lactobacillus* spp. in the three treatment groups for sorghum straw silage, while the *Alcaligenaceae* dominated the treatment group for extrusion puffing.

Linear discriminant analysis effect sizes (LEfSe) were used to evaluate the differences in microbial communities between treatments on sorghum straw and extrusion puffing sorghum straw (LDA score > 3.0). Figure 4a, b demonstrate the differences in microbial communities at the genus level after 30 and 60 days of silage for the different treatment groups, respectively. At 30 days of silage, the differential flora in the FM treatment

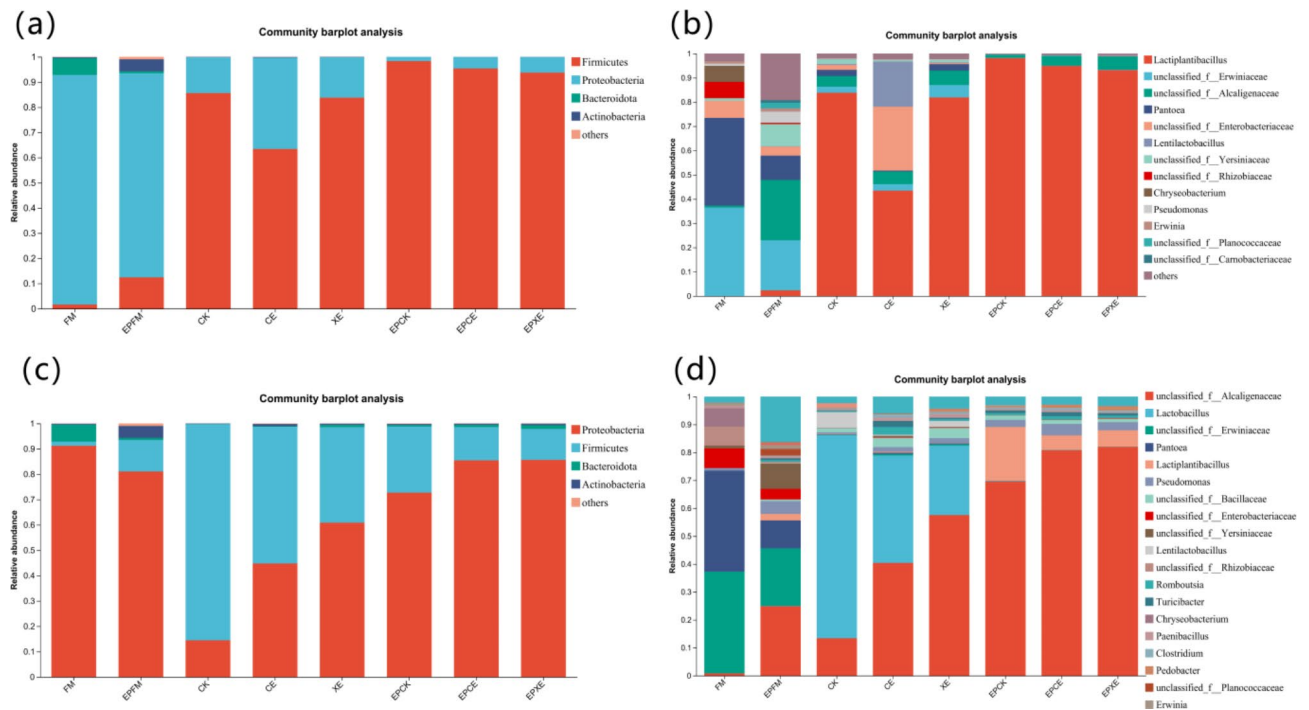


Fig. 2. Bacterial communities at the phylum and genus level for FM, EPFM, and silage. **a** Bacterial communities at the gate level for 30 d of silage. **b** Bacterial communities at the level of the 30-d silage genus. **c** Bacterial communities at the 60d silage gate level. **d** Bacterial communities at the 60 d silage level. FM, fresh material; EPFM, extrusion puffing material; CK, additive-free control; CE, cellulase; XE, xylanase; EPCK, additive-free control after extrusion puffing; EPCE, cellulase added after extrusion puffing; EPXE, xylanase added after extrusion puffing.

group consisted mainly of *Erwinia* and *Pantoea*, among others. While in EPFM, the differential flora was *Yersiniaceae*, *Pseudomonas*, *Pseudoxanthomonas*, and *Clostridium*. The EPCK treatment group significantly promoted *Lactiplantibacillus* content during silage, in contrast to the CK, EPCE, and EPXE groups which did not show significant differential flora. After 60 days of silage, the microbial community structure changed in each treatment group. *Lactobacillus* and *Lentilactobacillus* were more competitive in the CK group. The content of an OTU from an unclassified genus belonging to the family *Bacillales* was higher in the CE group. *Alcaligenaceae* mycorrhizal families were higher in the XE group. Two genera, *Lactiplantibacillus* and *Caldibacillus*, received better growth conditions in the EPCK treatment group. *Alcaligenaceae* and *Rhodococcus* were higher in the EPXE group. In contrast, the EPCE treatment group did not find significant differential flora after 60 days of silage.

Predicting metabolic pathways on the three levels

After PICRUST analysis, the six treatment groups exhibited different bacterial metabolic patterns based on the KEGG database, and Fig. 5 shows the metabolic pathways at 30d of silage. At the level of the first pathway (Fig. 5a), the relative abundance of metabolism was significantly higher than that of the other pathways, which were all above 70%, with the rest being environmental information processing and genetic information processing. Figure 5b shows the second pathway level, demonstrating the top 10 functions, of which six pathways belong to metabolism and the remaining four pathways belong to genetic information processing, environmental information processing, and cellular processes. Primary and stimulated metabolism showed a high relative abundance of metabolic pathways in treated sorghum straw compared to feedstock compared to feedstock. The tertiary metabolic pathways are shown in Fig. 5c, d. Figure 5c shows carbohydrate metabolism, where glycolysis/glycolysis breaks down glucose into pyruvate and produces a small amount of ATP through a series of enzymatic reactions. It can also be converted to glucose by non-sugar precursor substances such as lactic acid and glyceraldehyde-3-phosphate. Figure 5d shows the amino acid metabolism, Cysteine, and methionine metabolism. Glycine, serine, and threonine metabolism and Lysine biosynthesis were significantly increased in each treatment group after silage, which is involved in protein synthesis, energy metabolism, and other various physiological processes such as protein synthesis and energy metabolism. Figure 5e shows the metabolism of cofactor and vitamin pathways, with higher abundance in the CK, CE, and XE groups, and a decreasing trend in porphyrin and chlorophyll metabolism in the remaining groups (Fig. 5e). Metabolic changes in the two-component system of the signal transduction metabolic pathway were greatest after silage, probably reflecting the enhanced ability of the microorganisms to respond to external signals as well as the coordinated role of metabolic regulation (Fig. 5f).

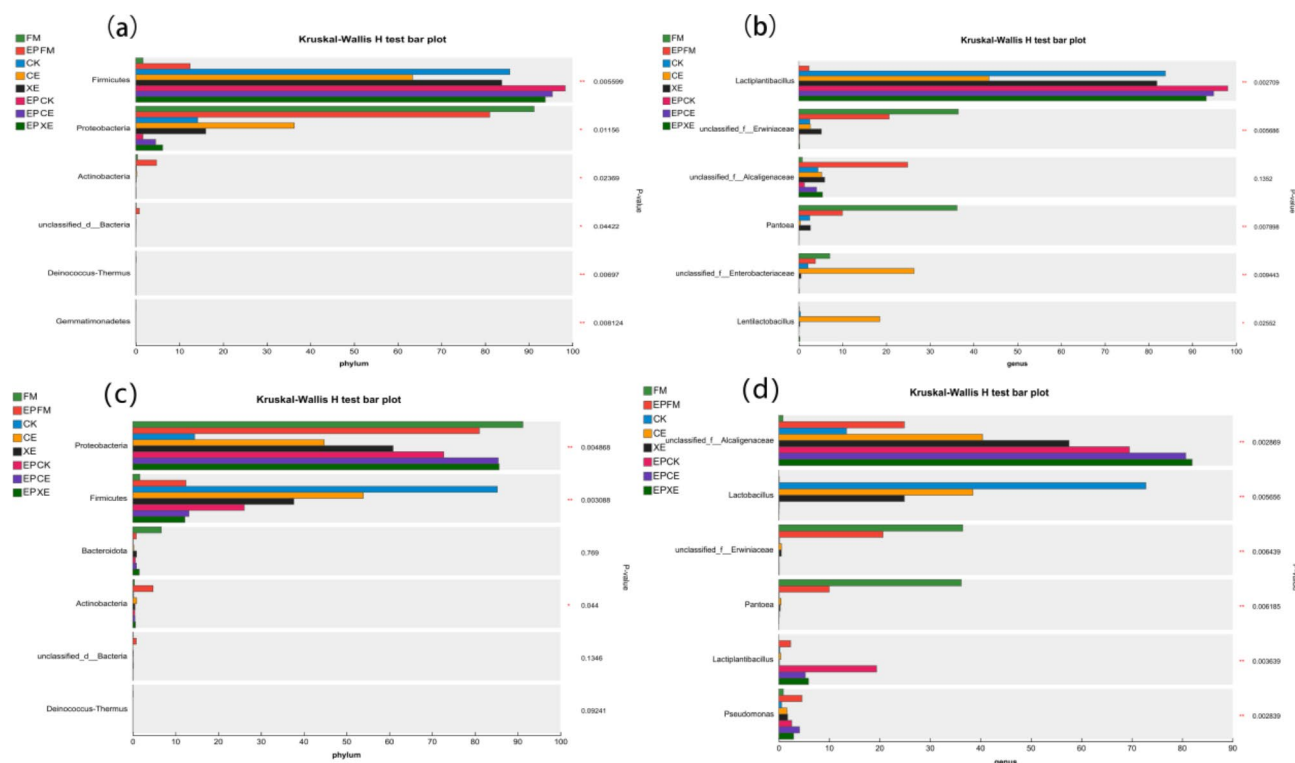


Fig. 3. Bacterial communities of FM, EPFM, and sorghum straw silage treatment groups. **a** One-way ANOVA bar graphs at the 30-day silage gate level. **b** One-way ANOVA bar graph for 30-day genus level of silage. **c** One-way ANOVA bar graphs at the 60-day gate level of silage. **d** One-way ANOVA bar graph for 60-day genus level of silage. FM, fresh material; EPFM, extrusion puffing material; CK, additive-free control; CE, cellulase; XE, xylanase; EPCK, additive-free control after extrusion puffing; EPCE, cellulase added after extrusion puffing; EPXE, xylanase added after extrusion puffing.

Figure 6 shows the metabolic pathways for 60d of silage. Primary and secondary metabolic pathways were similar to the 30d silage trend. As shown in Fig. 6c, the pathways of pyruvate metabolism, starch, and sucrose metabolism significantly ($p < 0.05$) increased in abundance with increasing silage time in all treatment groups. The metabolism of some amino acids showed an upward trend after 60d of silage, which had a positive effect on protein integrity and function. (Fig. 6d). The metabolism of porphyrins and chlorophyll was significantly ($p < 0.05$) boosted in the metabolism of cofactor and vitamin pathways after 60 d of silage (Fig. 6e). Signal transduction was significantly higher compared to 30 d of silage (Fig. 6f).

Scanning electron microscopy analysis

Figure 7a shows the scanning electron microscope image of the raw sorghum straw, the surface of the sorghum straw is smooth, with no obvious pores and faults, and its tissue and fiber structure is more complete. Figure 7b shows the scanning electron micrographs of sorghum straw after extrusion puffing, after treatment its sorghum straw surface became rougher, obvious faults appeared, the tissue on the outer surface began to undergo detachment, and the cell wall and complex fiber structure were destroyed.

The morphology of the samples was investigated using scanning electron microscopy. Figure 8(a) and 8(d) are scanning electron microscope (SEM) images of sorghum straw and extrusion puffing sorghum straw, respectively. As can be seen from Fig. 8b, c, the surface of sorghum straw after cellulase and xylanase treatments became roughened, and the surface of the enzymatically hydrolyzed sorghum straw particles left a lot of enzyme-dissolved pores, which indicated that the surface structure of the straw was destroyed by the enzyme treatments or new surface features were generated. The enzyme treatment successfully broke down some of the connections between the fiber bundles, making them easier to break down or utilize. In addition, there is significant fiber breakage and stripping, which alters the overall structure of the straw. However, sorghum straw was extrusion puffing to add enzymes for silage, and more holes, cracks, and looser fiber structure appeared on its surface, and the specific surface area of the straw was increased. Whereas, Fig. 8e, f show the sorghum straw with enzyme addition after extrusion puffing, which has a rougher surface with more pronounced faults and pores.

XRD analysis

XRD analysis was used to confirm the effect of enzyme addition and post-extrusion puffing fermentation on the structure as well as the crystallinity of sorghum straw. The XRD curves of FM and EPFM fibers are shown in Fig. 9. The characteristic peaks of sorghum straw were all significantly reduced after extrusion puffing. Characteristic peaks appeared at 2θ of 16.3° , 22.2° , 26.6° and 34.6° for the sorghum straw feedstock, indicating

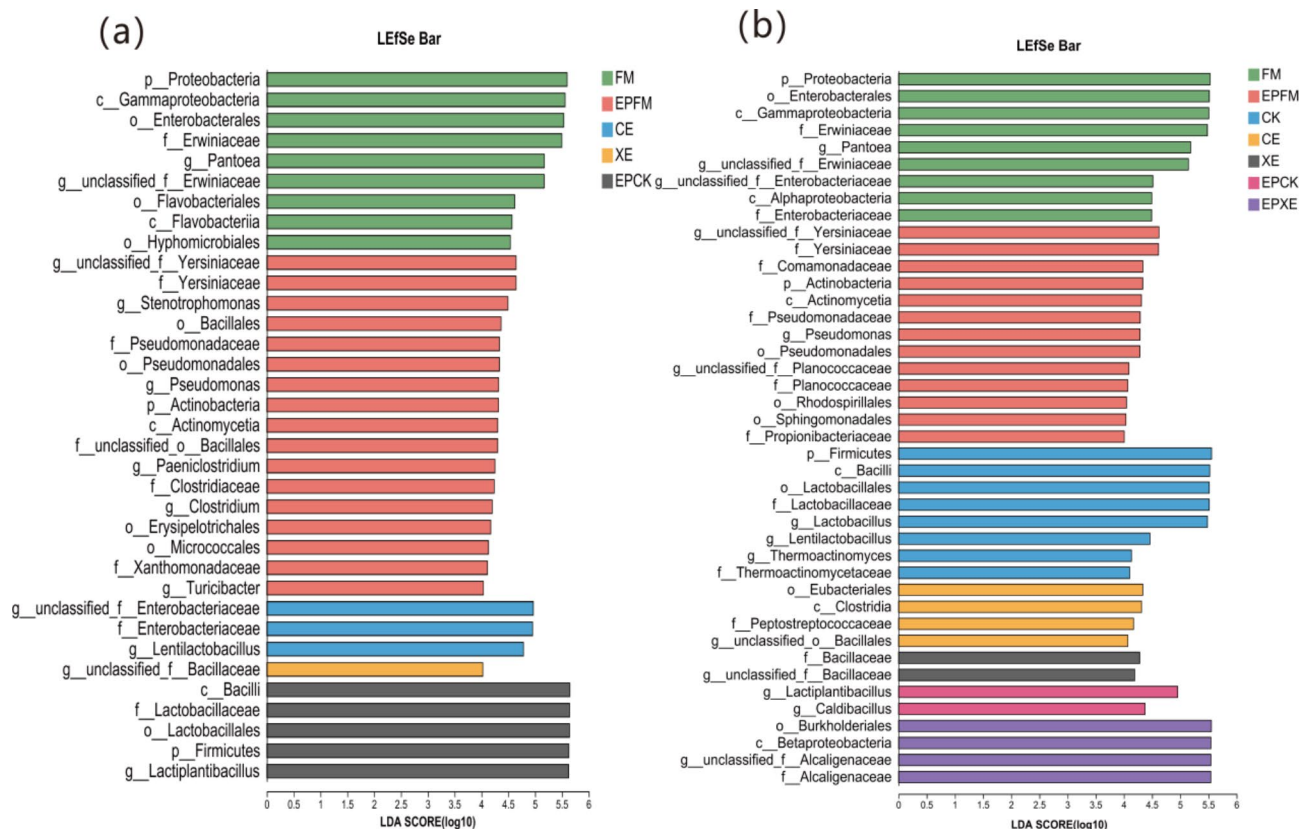


Fig. 4. Identifies groups that are significantly different between the six treatment groups. **a** 30-day silage community differences. **b** 60-day silage community differences. FM, fresh material; EPFM, extrusion puffing material; CK, additive-free control; CE, cellulase; XE, xylanase; EPCK, additive-free control after extrusion puffing; EPCE, cellulase added after extrusion puffing; EPXE, xylanase added after extrusion puffing.

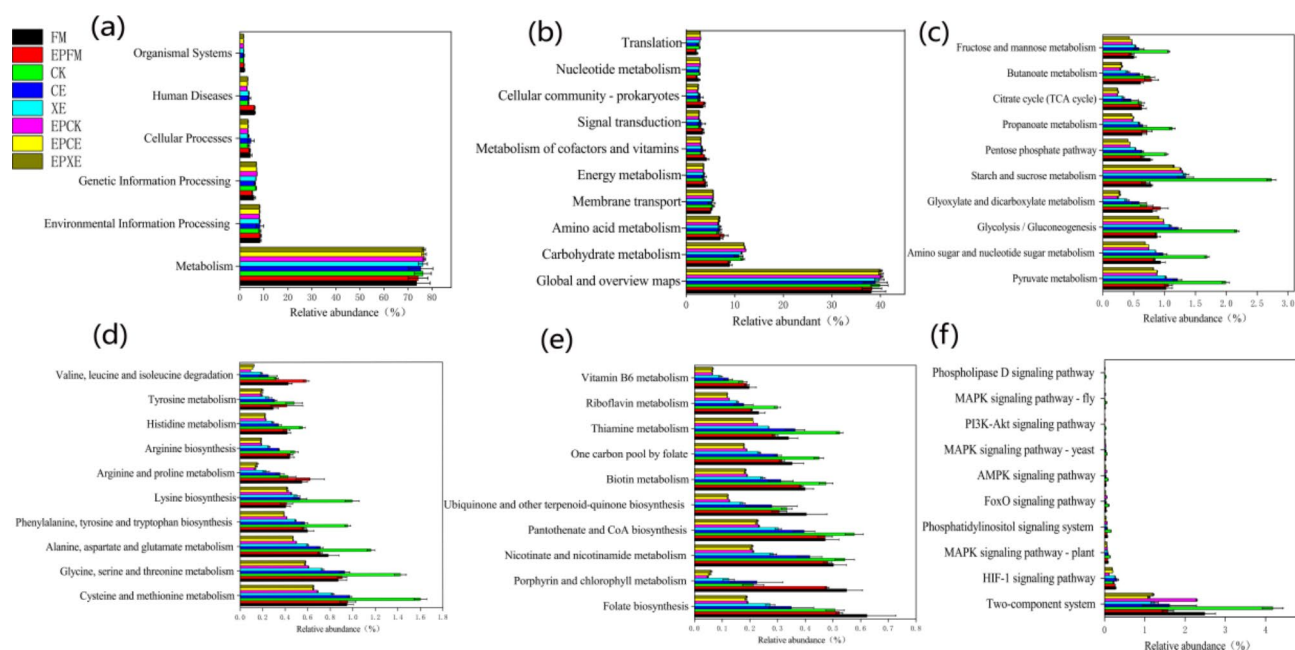


Fig. 5. Bar graphs showing the functional profiles of 16S rRNA gene predictions for fresh material and after 30 days of silage for the first **(a)** and second **(b)** pathway water, as well as the third pathway level for **(c)** carbohydrate metabolism, **(d)** amino acid metabolism, **(e)** cofactor and vitamin metabolism, and **(f)** signal transduction metabolism.

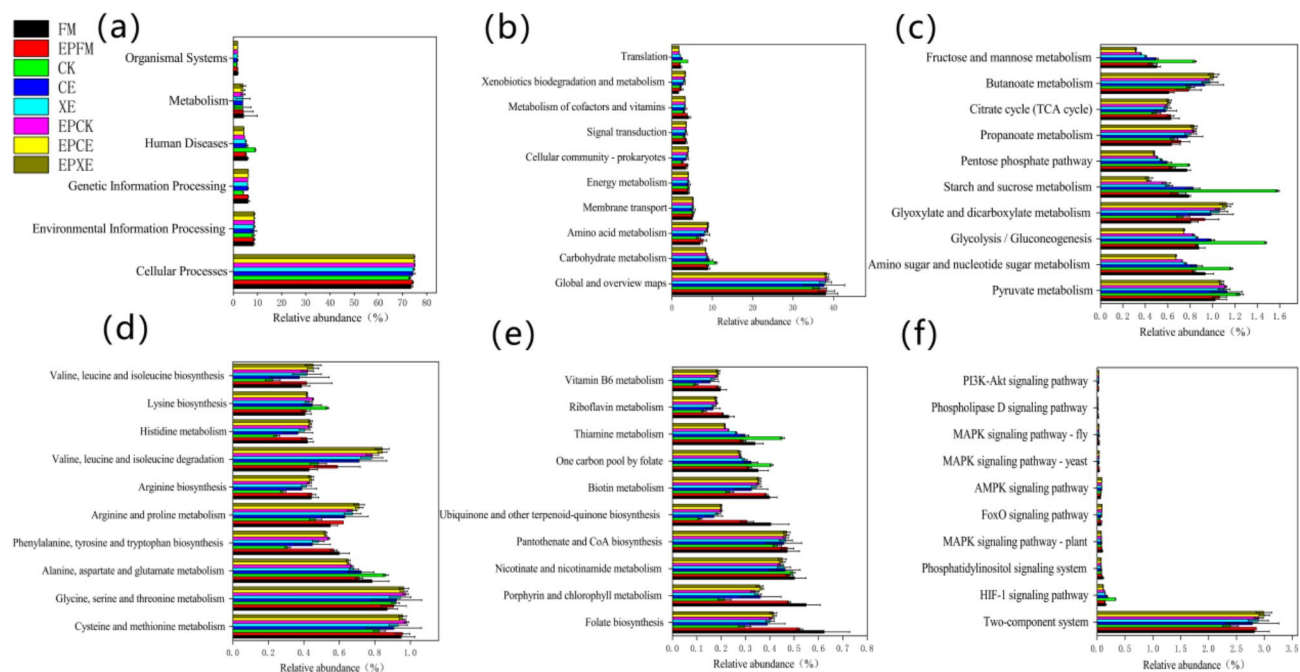


Fig. 6. Bar graphs showing the functional profiles of 16SrRNA gene predictions for fresh material and after 60 days of silage for the first (a) and second (b) pathway water, as well as the third pathway level for (c) carbohydrate metabolism, (d) amino acid metabolism, (e) cofactor and vitamin metabolism, and (f) signal transduction metabolism.

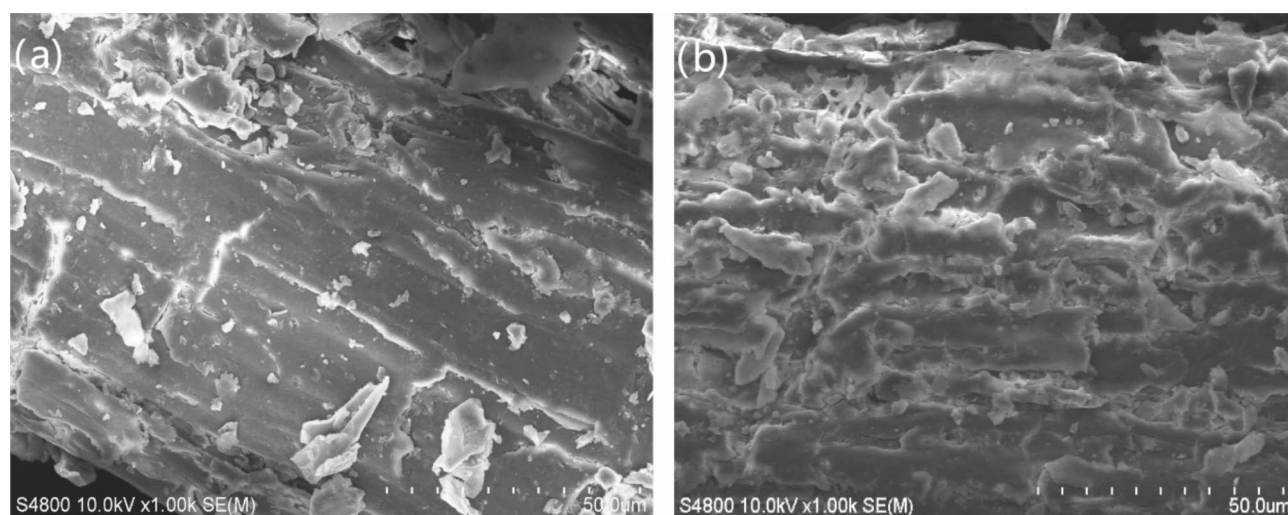


Fig. 7. Scanning electron microscope images of samples of raw sorghum straw and extrusion puffing sorghum straw feedstock. **a** FM, **b** EPFM.

that the cellulose in sorghum straw is a crystalline structure of type I cellulose. The substance corresponding to the strongest peak (200) is $C_{16}H_{14}O_5$ [14387-30-5], which occurs at 2θ equal to 22.2° , and the substance with the next highest peak (006) is C, which occurs at 2θ equal to 26.6° . The 111 characteristic peak has a broader diffraction peak in the range of 2θ of 14° to 18° , representing the diffraction peak of $C_6H_{12}O_6$ [93780-23-5]. Further cellulose crystallinity was calculated from the area of crystalline and non-crystalline zones and it was found that sorghum straw decreased from 27.8 to 13.8% after extrusion puffing. Reduced crystallinity is a very important indicator of reduced cellulose content or disrupted fiber structure.

The XRD results are shown in Fig. 10. In general, the addition of enzyme preparation and extrusion puffing treatment was able to degrade lignocellulose effectively. All diffractograms have three characteristic peaks at about 16.4° , 22.6° , and 26.5° , corresponding to the (111), (200), and (006) reflective surfaces of typical cellulose I structures, respectively. Compared to the control group, the crystallinity tended to decrease in all treatment

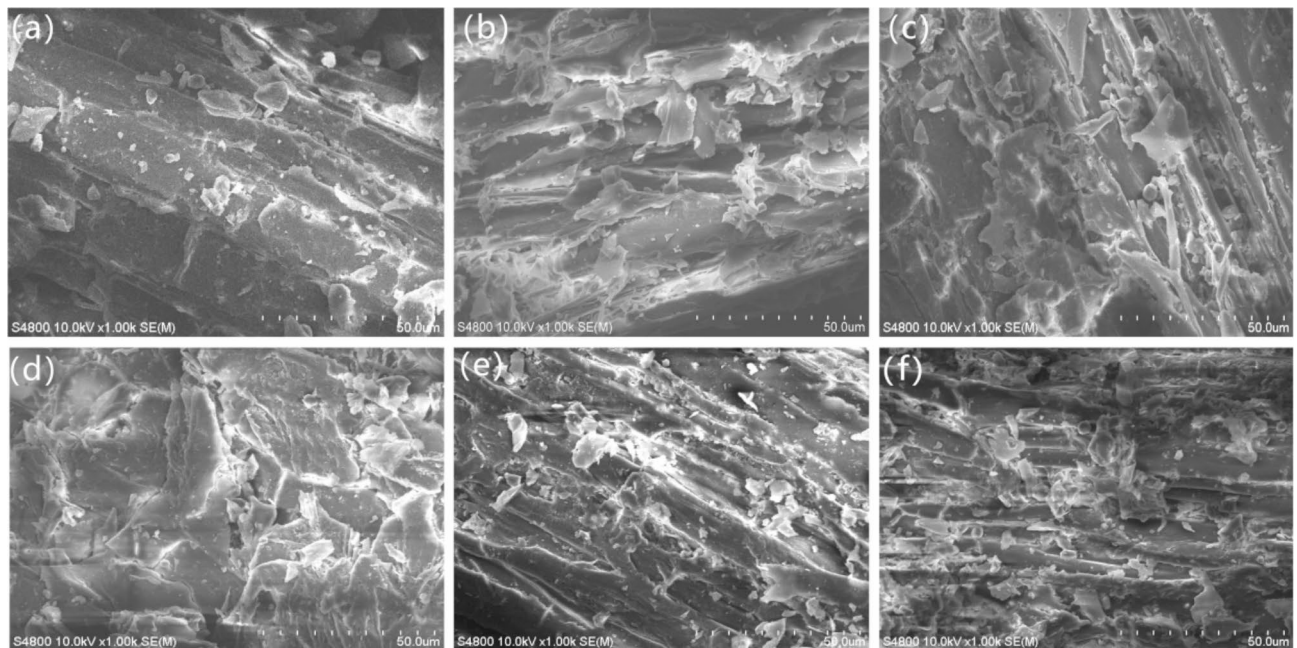


Fig. 8. Scanning electron microscope images of different sorghum straw samples after 60 days of silage. **a** CK, **b** CE, **c** XE, **d** EPCK, **e** EPCE, **f** EPXE.

groups. The cellulose crystallinity was further calculated from the area of crystalline and non-crystalline zones and the sorghum straw fermented for 30 days is shown in Figures (a) and (b) and the crystallinity indices calculated were 33.7%, 25.9%, and 22.9% for CK, CE, and XE, respectively. While the crystallization indices of EPCK, EPCE, and EPXE were 30%, 27.8%, and 26.8%, respectively. Also, it can be seen from the figure that the relative crystallinity decreased significantly after the addition of enzyme preparation. Figure 10c, d show the XRD plots of sorghum straw after 60 days of silage. The relative crystallinity of silage after enzyme addition and extrusion puffing also showed a decreasing trend, indicating that lignin and amorphous materials were removed in large quantities in the high-temperature environment of straw extrusion puffing.

FTIR analysis

The chemical changes in the sorghum straw samples were analyzed by FTIR method based on the position and shape of the absorption peaks determined by infrared spectroscopy. In Fig. 11, all the fluctuation curves associated with the cellulose, hemicellulose, and lignin fractions of sorghum straw were observed in the two main wave number intervals, $3500\text{--}2900\text{ cm}^{-1}$ and $1750\text{--}800\text{ cm}^{-1}$. The absorption peak at 3287 cm^{-1} corresponds to the O-H stretching vibration and the absorption peak at 2920 cm^{-1} is attributed to the stretching vibration of CH and CH_2 . 1721 , 1599 , 1366 , 1240 , 1034 , and 896 cm^{-1} , corresponding to C=O stretching vibrations (xylan), stretching vibrations of the benzene ring (lignin), CH_2 bending vibrations (lignin), CH bending vibrations (cellulose and hemicellulose), alkyl aryl ether-bonded C-O-C stretching vibrations (lignin), and C=O stretching vibrations (cellulose and hemicellulose), respectively. The absorption peaks located at 1034 and 896 cm^{-1} were interpreted as typical cellulose structures.

Discussion

Previous studies have shown that sorghum straw as a ruminant feed is ineffective for direct feeding and has relatively low digestibility due to its low crude protein content, high crude fiber, and poor palatability²⁴. Cellulases and xylanases have been widely used to improve silage fermentation quality and can effectively enhance its in vitro digestibility²⁵. High-quality silage pH should be ≤ 4.2 to ensure fermentation effect and nutritional value. The lowest pH in the cellulase addition treatment group in this study was attributed to the powerful effect of cellulase, which breaks down cellulose and provides more fermentable substrate for lactic acid bacteria, which in turn facilitates the cellulose degradation process. With the enhancement of lactic acid bacteria activity, the substrate was utilized for fermentation to produce acidic substances that inhibited the growth of harmful substances and ensured stable fermentation of silage in an anaerobic environment, which is in agreement with the previous findings of Li et al.⁷. As silage duration increased, unclassified *Alcaligenaceae* replaced *Lactobacilli* as the dominant genus at the later stages of silage through competitive growth, and pH increased in all treatment groups, probably due to increased degradation of *Alcaligenaceae* and their production of alkaline metabolites²⁶. The extrusion expansion technology, by altering the chemical structure of biomass, disrupts the cell walls, increases surface area and porosity²⁷, thereby facilitating enzymatic digestibility, cellulose degradation, protein transformation, and the growth and lactic acid production of *Lactobacilli* and other microbes²⁸. This elucidates the observed increase in the contents of LA and AA over time in the XE and EPXE groups. With the extension of silage time, the microbial metabolic activity during fermentation is enhanced, leading to a faster rate of

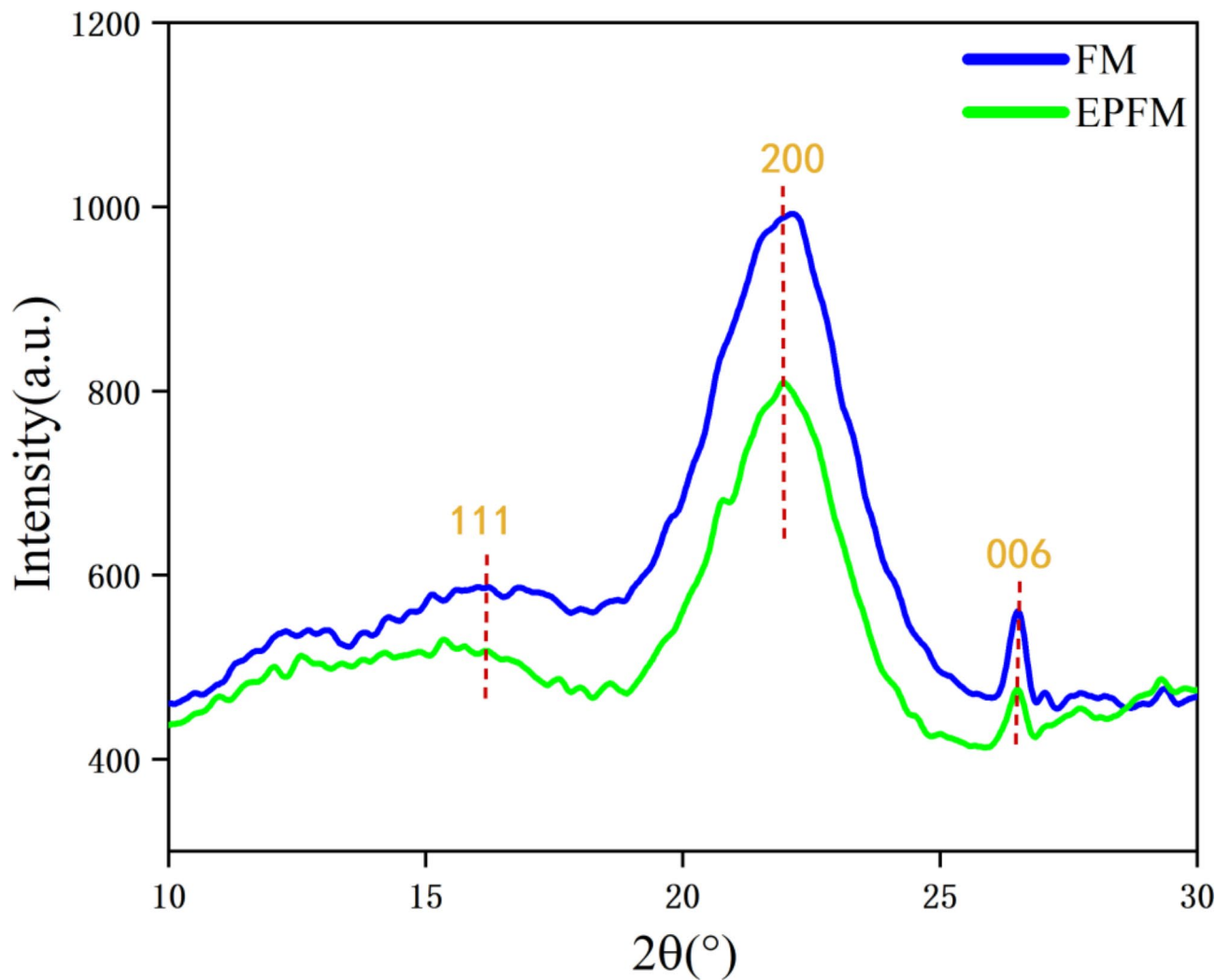


Fig. 9. XRD patterns of raw sorghum stalks and extrusion puffing sorghum stalks. **a** FM, **b** EPFM.

production and accumulation of $\text{NH}_3\text{-N}$ and an increase in its content²⁹. However, the specific effect of extrusion puffing on $\text{NH}_3\text{-N}$ content needs further study.

Chemical composition analysis showed that the addition of enzyme preparation and extrusion puffing treatment had a significant effect on the fermentation quality of sorghum straw silage. Among them, all treatment groups showed a significant increase in DM content at the late silage stage, which indicated that both extrusion puffing and enzyme treatment had an effect on DM content³⁰. As the ensiling period extends, the content of CP increases, possibly related to changes in the microbial community structure. *Alcaligenaceae* gradually becoming predominant in the later stages of silage, possess significant protein-synthesizing capabilities³¹. The CP content also tended to increase with the addition of enzymes, which may be attributed to the lowering of pH due to the production of acids such as lactic acid bacteria with the addition of cellulase and xylanase, which help to protect proteins from catabolism and metabolism³². NDF and ADF content are important indicators for assessing rumen digestive efficiency. When NDF levels are high, it makes rumen digestion more difficult. On the contrary, lower NDF levels help reduce the burden of rumen digestion and promote feed digestion and absorption. Cellulase and xylanase in silage significantly reduce NDF and ADF content by efficiently hydrolyzing cellulose and hemicellulose in the plant cell wall, thus enhancing the feed's nutritive value and rumen digestive efficiency³³. However, the NDF and ADF content of silage, although increased after the extrusion puffing treatment, leads to a decrease in digestibility, which is similar to that of the R. J. Consistent with the study by Gaebe et al.³⁴. Enzymes can promote the decomposition of carbohydrates in silage, especially the degradation of WSC. Lactic acid bacteria dominate the microorganisms when the silage is 30 days old, and WSC is the main substrate for lactic acid bacteria fermentation, its content directly affects the fermentation rate and quality of silage, and the content of WSC in each treatment group can meet the requirements of high-quality silage fermented feed. However, the decrease in WSC content with increasing silage time may be because silage is fermented in an anaerobic environment, and microorganisms such as lactic acid bacteria utilize the WSC in silage for growth and metabolism. Also, the extrusion puffing treatment had a significant effect on the WSC content, related to the

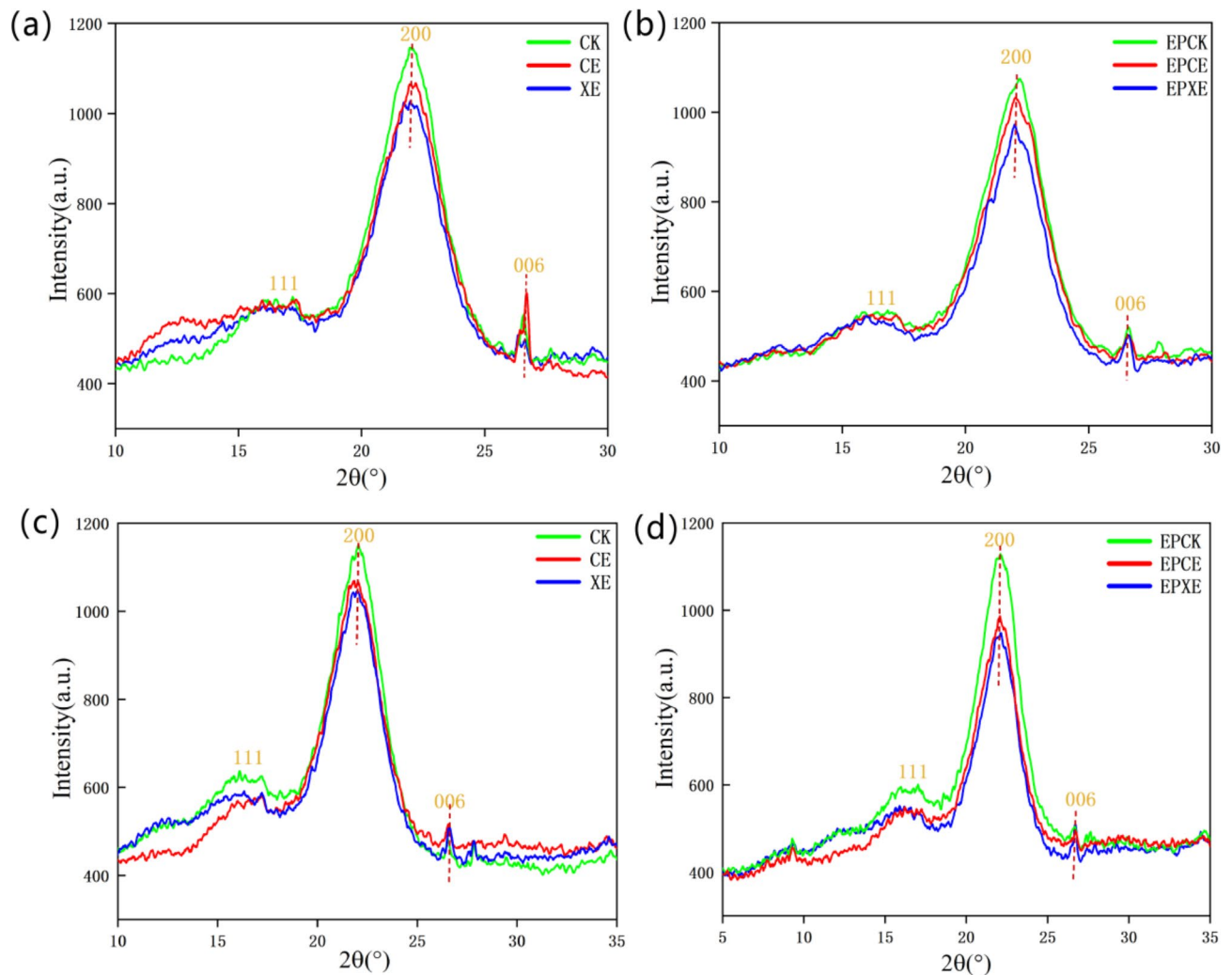


Fig. 10. XRD patterns of different treatment groups of sorghum stalks: **a** Sorghum stalk silage after 30 days, **b** extrusion puffing sorghum stalk silage after 30 days, **c** sorghum stalk silage after 60 days, **d** extrusion puffing sorghum stalk silage after 60 days.

disruption of the lignin sealing structure under high-temperature conditions as well as the efficient hydrolysis of cellulose or hemicellulose to oligosaccharides or monosaccharides under acidic conditions.

The addition of enzymes and the extrusion puffing treatment significantly affected the microbial community structure during the silage process. In this study, it was found that the Ace, Chao1, and Shannon indices of each treatment group tended to increase with the increase of silage time, especially, the treatment group with enzyme addition significantly decreased the Simpson index during the two periods of silage, which may be because enzymes provide more soluble sugars for lactic acid bacteria, produce more lactic acid to lower the pH of the feeds and inhibit the undesirable microbial proliferation, thus reducing bacterial diversity³⁵, which is consistent with the findings of Xiong et al.³⁶. However, the indices were significantly lower in the extrusion puffing treatment group, which indicated that the extrusion puffing treatment might have promoted the release and utilization of nutrients by changing the structure of the straw, but at the same time, it also intensified the competition among the microorganisms, which led to the rapid proliferation of certain dominant strains, while the microbial diversity was reduced accordingly. In addition, PCoA plots demonstrated differences in bacterial communities between raw material and silage samples, suggesting that the silage process reconfigures the bacterial community.

Bacterial communities are closely related to silage quality, and bacterial interactions during the silage process are critical. The present study revealed that the Proteobacteria phylum was more abundant at the beginning of silage, but Firmicutes phylum bacteria gradually dominated with the extension of silage time. This phenomenon is mainly attributed to the fact that most of the bacteria involved in lactic acid fermentation belong to the Firmicutes, which produce lactic acid by fermentation, contributing to the preservation of silage³⁷. After 30 d of silage, the proportion of Firmicutes increased after extrusion puffing treatments. However, at 60 days of silage, the proportion of Proteobacteria increased significantly in all treatment groups, which may be related to the destruction of cellulose structure and inhibition of microbial growth by extrusion. Proteobacteria is an

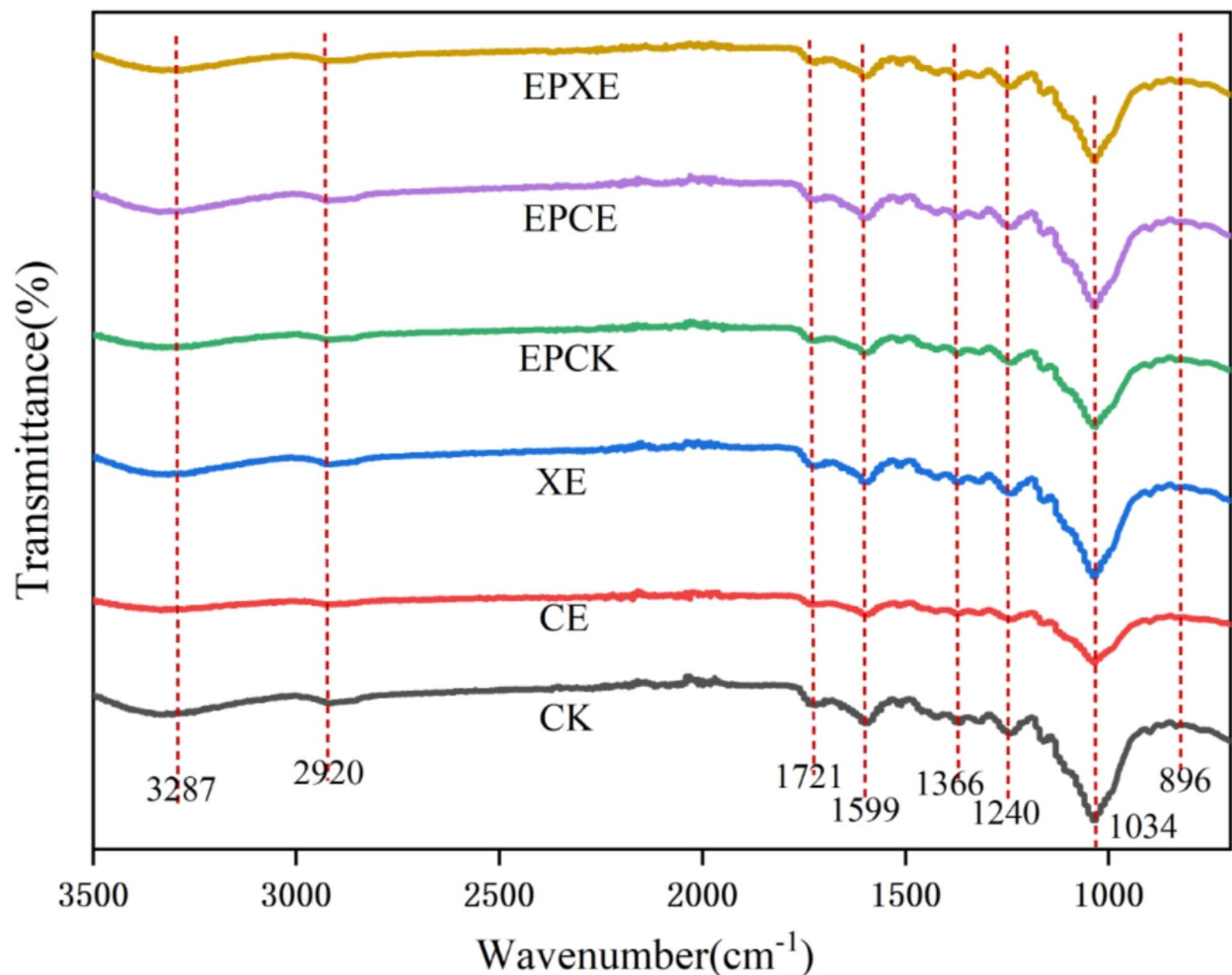


Fig. 11. FTIR spectra of different treatment groups of sorghum stalks.

important phylum within the bacterial domain in which *Alcaligenaceae* can produce ammonia and carbon dioxide through specific metabolic pathways. Ammonia, as a strong alkaline, neutralizes organic acids such as lactic acid produced by lactic acid bacteria fermentation in silage, thus raising the pH of the environment. In addition, microorganisms may release other alkaline substances as they break down organic material in silage, further affecting the pH. In addition, there may be interspecies competition between *Lactobacilli*, *Alcaligenaceae*, and enzymes. *Lactobacilli* inhibit the growth of other microorganisms by producing organic acids through fermentation and lowering the pH. In contrast, *Alcaligenaceae* may neutralize lactic acid by producing alkaline substances, weakening the competitive advantage of lactic acid bacteria³⁸. Also, the addition of enzymes to the extruded puffing treatment may accelerate the decomposition and utilization of fermentable sugars, which both promote the growth of *lactobacilli* and may lead to a decrease in fermentable sugars, which affects the growth of other microorganisms and makes *Proteobacteria* more vigorous.

Enzymatic and extrusion puffing treatments also had an effect on the colony structure of sorghum straw at the bacterial genus level. For 30 days of silage, *Lactiplantibacillus* was the main dominant genus, whose acid-producing capacity contributes to the formation of high-quality silage. However, the lower content of *Lactiplantibacillus* in the CE group may be due to the loss of cellulase activity at lower pH, resulting in limited WSC production. In contrast, the XE group had higher levels, which may be due to the strong breakdown of hemicellulose by xylanase. *Enterobacteriaceae* are non-pathogenic during the fermentation of silage, but their growth is detrimental, as they compete with lactic acid bacteria for WSC while producing $\text{NH}_3\text{-N}$ and slowing down the decrease in pH, thus affecting the quality of silage fermentation³⁹. For 60 d of silage, the content of *Alcaligenaceae* increased dramatically in all treatment groups, most significantly with the addition of enzymes and extrusion puffing treatments. *Alcaligenaceae* genera do not directly hydrolyze cellulose, but may indirectly utilize cellulose decomposition to produce intermediate products such as oligosaccharides through synergistic interactions with other microorganisms. During extrusion puffing sorghum straw, cellulose may be partially degraded to these intermediates, thus providing conditions for the growth of *Alcaligenaceae* genera⁴⁰. The extrusion puffing treatment changes the microbial community structure, effectively reduces the content of NDF and ADF, creates more favorable conditions for microbial growth and reproduction, and provides

more abundant fermentable sugar substrates. These microorganisms utilize sugars for metabolism to produce more organic acids, further improving the nutritional value and digestibility of the feed. In silage, cellulase and xylanase are added and treated with extrusion puffing, and their synergistic effects are induced by a variety of microorganisms⁴¹. Firstly, adding cellulase and xylanase, can promote the growth of beneficial bacteria and improve the quality of silage fermentation. Secondly, *Alcaligenaceae* usually produce alkaline substances, and these *Alcaligenaceae* have special metabolic pathways or enzyme systems that enable them to efficiently utilize soluble sugars and produce acidic substances in the fermentation environment⁴². This conversion process is positive for the fermentation quality of silage, as it further inhibits the growth of harmful microorganisms and promotes the preservation of the feed and the maintenance of its nutritional value. The extrusion puffing treatment disrupts plant cell walls, releasing more nutrients and promoting the growth and fermentation of *Alcaligenaceae*⁴³. In addition, other microorganisms such as butyric acid bacteria are present in silage. These microorganisms may grow and multiply under specific environmental conditions and play a role in the silage fermentation process⁴⁴. However, the growth and action of these microorganisms were relatively weak with the addition of cellulase and xylanase and extrusion puffing treatment. In summary, the addition of cellulase and xylanase in the silage process and the combination of extrusion puffing treatment significantly enhanced the fermentation quality and nutritional value of silage by promoting the growth and fermentation of beneficial microorganisms such as lactic acid bacteria and *Alcaligenaceae*⁴⁵. The synergistic effect of these microorganisms not only enhances the safety of silage but also effectively improves its utilization efficiency, which provides a strong guarantee for the healthy growth and production performance of animals. Therefore, this treatment has a significant positive effect on optimizing silage quality.

Predictions of bacterial community functioning allow a more detailed understanding of the possible mechanisms influencing the dynamics of silage quality. However, it should be noted that this type of analysis is inferred from 16S rRNA genomics and does not directly measure metabolic pathways or enzymes. This study investigated the effects of enzyme and extrusion puffing treatments on the functional spectrum of bacterial communities in sorghum straw silage at different silage times. The high abundance of carbohydrate and amino acid metabolism in primary and secondary metabolism, and the high activity and metabolism of microorganisms reflect the high activity and capacity of the bacterial community.

On the tertiary metabolic pathways, carbohydrate metabolism, amino acid metabolism, cofactor and vitamin metabolism, and signal transduction metabolism are the major metabolic pathways. Carbohydrate metabolic pathways increased dramatically in abundance with silage time and consisted of three key components: pyruvate conversion, gluconeogenesis, and glycolysis. During the silage process, the cells are exposed to a hypoxic environment due to limited oxygen supply, at which time the glycolytic pathway becomes the main way for the cells to obtain energy. By promoting enzymatic reactions such as the addition of enzymes, glucose is progressively broken down to pyruvate and in the process, a small but vital amount of ATP is released to provide the necessary energy support for the survival and metabolic activities of the microorganisms in the silage⁴⁶. It is important for improving the quality and storage stability of silage.

The metabolic pathways of most amino acids were inhibited during the initial period of silage, which is consistent with the low $\text{NH}_3\text{-N}$ content of silage. The relative abundance of amino acid metabolism increased in all treatment groups at 60d of silage, which was consistent with the results of increased pH and increased *Alcaligenaceae*. It is possible that *Alcaligenaceae* utilizes amino acids as a source of energy or performs other metabolic activities that promote the metabolism of amino acids. Thus, the increase in *Alcaligenaceae* is an important factor in the enhanced metabolic activity of amino acids during the late silage period. Metabolic pathways included in amino acid metabolism are involved in a variety of physiological processes such as protein synthesis and energy metabolism⁴⁷. Analysis of metabolic pathways revealed how amino acids are converted into other substances such as proteins and a variety of metabolic intermediates within sorghum straw. These metabolites are performing functions such as structural support and enzyme catalysis.

Deficiencies of certain cofactors or vitamins may lead to metabolic disorders and disease development, while excessive intake may produce toxic effects. Cofactor and vitamin metabolism levels were higher in FM and EPFM, but their metabolism decreased after silage, probably due to the destruction of the fiber structure and vitamin production caused by the addition of enzymes and extrusion puffing treatment. Signaling pathways transmit information through specific signaling molecules, which in turn regulate the physiological functions of cells or tissues, and may lead to disruption of cellular or tissue physiological functions, which may in turn lead to disease. For 60d of silage, the increase in signal transduction abundance may be related to the increase in pH and the decrease in organic acid content.

Scanning the morphology of the sorghum straw samples reveals that the surface of the enzyme-treated sorghum straw is rougher, indicating that the surface structure of the straw has been destroyed by the enzyme treatment or new surface features have been created. There was also significant fiber breakage and stripping, which are usually due to enzyme action resulting in weakened or broken connections between fibers, thus affecting the overall structure of the straw. The enzyme treatment successfully broke down some of the connections between the fiber bundles, making them easier to break down or utilize. The smaller diameter of fiber bundles and the appearance of holes or gaps inside are the results of the change in the internal structure of fibers after cellulase and xylanase decompose cellulose and hemicellulose in straw. Sorghum straw is more loosely structured after extrusion puffing, and the loose structure is more favorable for enzyme action and subsequent biodegradation processes⁴⁸, and the specific surface area of the straw has increased, meaning that sorghum straw is more likely to come into contact with enzymes or other microorganisms, which can improve its biodegradability or nutritional value. Thus the structural changes were most pronounced in the group of treatments where enzymes were added after extrusion puffing. Whereas at 60 days of silage, *Alcaligenaceae* become dominant and the increase of *Alcaligenaceae* may form some special structures or textures on the surface of sorghum straw. The fungus may

adhere to the surface of the straw, forming a biofilm or colony, or the fungus may erode the surface of the straw, therefore making it rough and uneven.

In sorghum straw, cellulose, hemicellulose, and lignin together form its structural basis. Most of these biomacromolecules exist in an amorphous or partially crystalline state and thus show broadened peaks or diffuse backgrounds in the XRD plots. By XRD analysis, 111 diffraction peaks corresponding to $C_6H_{12}O_6$ (glucose) were observed, suggesting that glucose may be present in the straw in the crystalline state. After 60 days of silage, the intensity of this diffraction peak decreased significantly in the treatment groups with enzyme addition, which may be attributed to the fact that the enzyme and extrusion puffing treatments disrupted the cellulose, hemicellulose, and lignin structures in the sorghum stover, which in turn affected the crystallization of glucose. Lignin is a series of polymers of phenylpropane units, consisting of phenylpropane units connected by ether and C-C bonds, which contain a large number of carbon atoms and a small number of hydrogen and oxygen atoms, so the substance at the highest diffraction peak (200) is $C_{16}H_{14}O_5$, and the degree of crystallinity of all the treatments was significantly reduced here compared with the control group, and it was more pronounced in the 60th day of silage, which is in line with the results of the crystallinity index, which indicated that the crystal structure of sorghum straw samples was disrupted during the removal of cellulose, hemicellulose and lignin content⁴⁹.

Enzyme treatments significantly reduced the intensity of absorption peaks associated with cellulose and hemicellulose, suggesting that the enzymes effectively degraded these complex carbohydrates. The extrusion puffing treatment further changed the structure of the straw, resulting in clearer and more pronounced characteristic peaks in the FTIR pattern. Compared to the control CK and EPCK, the other treatment groups tended to level off in all wave number bands, a result that suggests that cellulose, hemicellulose, and lignin are in a cleavage state. These changes not only reflect the degradation and structural changes of biomolecules in straw but also predict the improvement of its biodegradability and availability. These findings are important for improving energy utilization efficiency and environmental applications of sorghum straw. In addition, the replacement of *Lactobacilli* by *Alcaligenaceae* as the dominant bacteria on 60d of sorghum straw silage may also change some of the chemical composition or chemical bond types in the straw. If certain metabolites (e.g., polysaccharides, proteins, etc.) are produced by *Alcaligenaceae*, these products may produce new absorption peaks on the FTIR plot. Or if the *Alcaligenaceae* alters certain chemical bonds (e.g., ester bonds, hydroxyl groups, etc.) in the straw, the corresponding absorption peaks will also be changed, which requires further observation and research.

Conclusion

This study investigated the effect of enzyme preparation and extrusion puffing on sorghum straw silage fermentation. The results showed that both significantly improved silage quality, such as pH regulation, enhanced digestibility, and nutritive value, by modulating the bacterial community structure, especially the enhancement of the dominance of *Alcaligenaceae*. Xylanase is more effective than cellulase, which can effectively degrade xylan, enhance soluble sugar content, and promote microbial growth. Extrusion puffing enhances the enzymatic efficiency by changing the physical structure of the straw. The enzyme preparation significantly promoted cellulose and hemicellulose decomposition, which was verified by scanning electron microscopy observation of fiber bundle refinement and increased pore size. XRD and FTIR analyses confirmed that enzyme treatment induced the structural transformation of complex carbohydrates to produce more digestible soluble sugars. In summary, enzyme preparation and extrusion puffing treatment can effectively optimize the sorghum straw silage fermentation process.

Data availability

The datasets generated and/or analysed during the current study are available in the NCBI repository, study record number PRJNA1153972.

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Author contributions

Y.X.S. wrote the main manuscript text. M.J.L. and Y.C.L. review and revision. P.J.S. and B.C.B. review and data curation. R.Y.B. Investigation and resources. T.Y.L. and K.S. Project administration and funding acquisition. All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Additional information

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